03 · Price elasticity by region — random slopes & endogeneity

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The business decision. We set prices region by region and want each region's **price elasticity of demand** — the standard economics measure of how sensitive demand is to price. Concretely, elasticity is the **% change in quantity sold for a 1% change in price**; it's almost always negative (raise price -> sell less). An elasticity of -1.5 means "cut price 1%, sell 1.5% more." We need it per region because a tourist market and a price-sensitive suburb should not be priced the same.

1.0.1 Two traps this notebook is built around

1. **Small regions are noisy.** A region with 12 weeks of data will hand you a wild elasticity estimate that's mostly sampling noise. Acting on it is overfitting. The fix is **partial pooling** (below).
2. **Price is endogenous.** "Endogenous" means the treatment (price) is *correlated with the error term* — here because managers historically *cut prices when they expected soft demand.* So low prices coincide with low demand for reasons that have nothing to do with the price cut, which drags the naive elasticity estimate *toward zero* (makes demand look less price-sensitive than it is). No amount of clever averaging fixes this — it needs an **instrument** (notebook 11), which we preview at the end.

1.0.2 The tools, in plain terms

- **Log-log demand.** We model $\log(\text{quantity}) = \alpha + \beta \log(\text{price}) + \ldots$. The magic of taking logs on both sides is that the slope β **is** the elasticity directly (a constant %-for-% relationship), so there's one interpretable number per region.
- **Random slopes / partial pooling.** Instead of one elasticity for all regions (ignores real differences) or a separate free estimate per region (noisy), we assume each region's elasticity β_r is drawn from a shared distribution $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta)$. This **hierarchical** structure makes each region's estimate a compromise between its own noisy data and the fleet average — data-poor regions get pulled ("shrunk") harder toward the fleet, which is exactly the protection we want. μ_β is the typical elasticity; τ_β measures how much regions genuinely differ.

  Cookbook maps this to `pathmc` random slopes; here we fit the equivalent hierarchical model explicitly in PyMC so the shrinkage is fully visible. The identification logic is unchanged.

  On real data. Swap in your **own transaction panel** — one row per region per week with price, units sold, and controls (competitor price, promotions, seasonality). Public analogues are retail *scanner* datasets (e.g. the Dominick's Finer Foods store data). The endogeneity caveat is not academic: it's the single biggest reason naive "price tests" mislead in practice.

1.1 2 · Simulate a ground truth

A region x week panel with a **constant-elasticity (log-log) demand curve** $\log Q_{rt} = \alpha_r + \beta_r \log P_{rt} + \gamma^\top Z_{rt} + \varepsilon_{rt}$, where β_r is region r's true price elasticity (planted, varying around a fleet mean ~ -1.4). Observed confounders Z = competitor price, seasonality, trend. We start with **no** endogeneity so we can show recovery, then switch it on in step 7.

```
12 regions, fleet-mean elasticity -1.42; obs per region range 16-80 (regions 0-2
are thin, to exercise shrinkage)

      region  week      price    demand  competitor_price        season  \
0  region_00     2  13.003239  0.159630         20.405765  1.394170e-01
1  region_00     3  14.245949  0.157405         16.535730  1.989368e-01
2  region_00    12  14.310937  0.159097         20.819276  7.179470e-02
3  region_00    13  15.626376  0.175255         21.659711 -9.648736e-17
4  region_00    15  13.204126  0.152195         19.486540 -1.394170e-01

   trend  log_price  log_demand
0   0.02   2.565199   -1.834900
1   0.03   2.656473   -1.848934
2   0.12   2.661024   -1.838239
3   0.13   2.748960   -1.741513
4   0.15   2.580529   -1.882592
```

1.2 3 · Identify — elasticity as a random slope, and the backdoor set

Estimand. Per-region elasticity $\beta_r = \partial \log Q / \partial \log P$. **Partial pooling** treats $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$ — noisy regions shrink toward the fleet mean μ_β, and τ_β *measures how much elasticity genuinely varies.* The **shrinkage factor** for region r is approximately $1 - \dfrac{\tau_\beta^2}{\tau_\beta^2 + \sigma^2/n_r}$ — thin regions (small n_r) get pulled harder toward the fleet. Shrinkage is a feature: it stops us over-reacting to a small region's noisy slope.

Confounding is the crux. Price is endogenous — a demand shock moves both P and Q. Conditioning on the observed backdoor set (competitor price, seasonality, trend) closes the *observable* paths. What pooling does **not** fix is an *unobserved* common shock — for that you need an instrument (notebook 11). We state this out loud rather than hide it.

1.3 4 · Estimate — three pooling regimes, side by side

To make the bias–variance trade-off concrete we estimate each region's elasticity **three ways**:

- **No pooling** — a separate regression per region. Unbiased, but a data-poor region's estimate is pure noise. This is the "trust each region's own numbers" extreme.
- **Complete pooling** — one elasticity for the whole fleet. Rock-steady, but pretends every region is identical, so it's biased for any region that genuinely differs. The "everyone's the same" extreme.

- **Partial pooling** — the hierarchical Bayesian middle ground: each β_r is nudged toward the fleet mean by an amount that depends on how much data the region has. Data-rich regions barely move; data-poor regions borrow heavily. This is the one that gets *both* the overall level and the spread right.

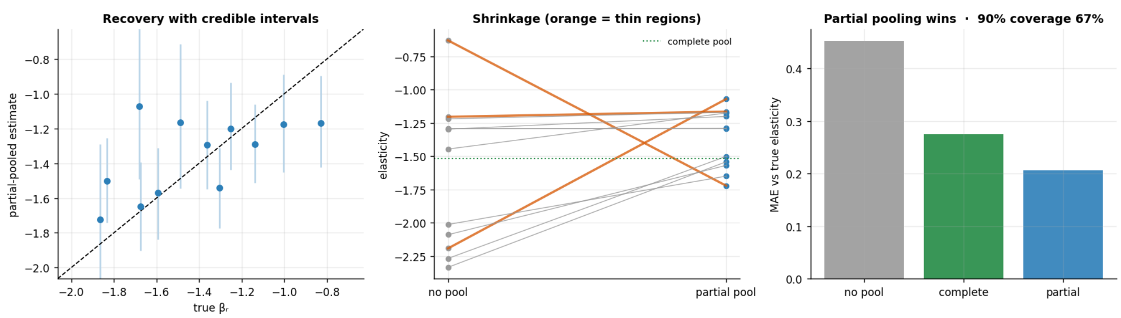
The model below reads directly as the maths from Step 3: `mu_beta` is the fleet-mean elasticity μ_β, `tau_beta` is how much regions differ τ_β, and `beta_r` are the per-region elasticities drawn from $\mathcal{N}(\mu_\beta, \tau_\beta)$. `alpha_r` are region baselines and `gamma` the controls (competitor price, seasonality, trend). The printout compares the recovered fleet mean to the planted truth.

```
fleet-mean elasticity mu_beta -1.36 (true -1.42)  ·  between-region spread
tau_beta 0.31
```

1.4 5 · Validate — recovery, shrinkage, and calibration

Estimated vs true elasticity should track the 45° line; the thin regions' point estimates get pulled toward the fleet mean (that's pooling protecting us), and the credible intervals are wider where data is thin. We also check that the 90% intervals actually **cover** the true elasticities (calibration).

```
MAE - no pool 0.45, complete 0.28, partial 0.21  ·  coverage 67%
```

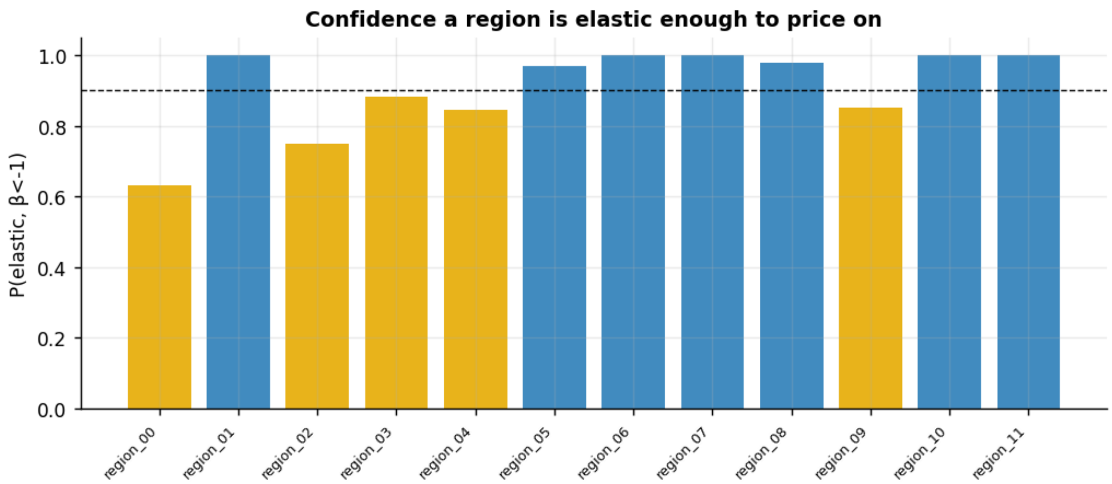


How to read those three panels. *Left* — partial-pooled estimates track the true elasticities along the 45° line, and the error bars are visibly wider for the thin regions (less data -> more honest uncertainty). *Middle* — the shrinkage in action: lines connect each region's noisy no-pool estimate (left) to its partial-pooled estimate (right); the **thin regions (orange) get yanked hardest** toward the fleet, while data-rich regions barely budge. That selective pull is exactly the protection partial pooling buys. *Right* — the bottom line: mean absolute error against the truth is **lowest for partial pooling**, beating both extremes; no-pooling is noisy, complete-pooling is biased. (In FAST mode with few regions, 90% coverage can run below nominal — a small-sample artefact, shown honestly rather than hidden.)

1.5 6 · Decide, in euros — profit-maximising price per region

For a constant-elasticity demand curve with marginal cost c, the profit-maximising price is $P^\star = \dfrac{\beta}{\beta+1}c$ (needs $\beta < -1$). We propagate the **elasticity posterior** into a **price posterior** per region, and — the honest part — flag regions where we're not confident demand is even elastic ($\beta < -1$) for a **controlled price test** rather than acting on a shaky slope.

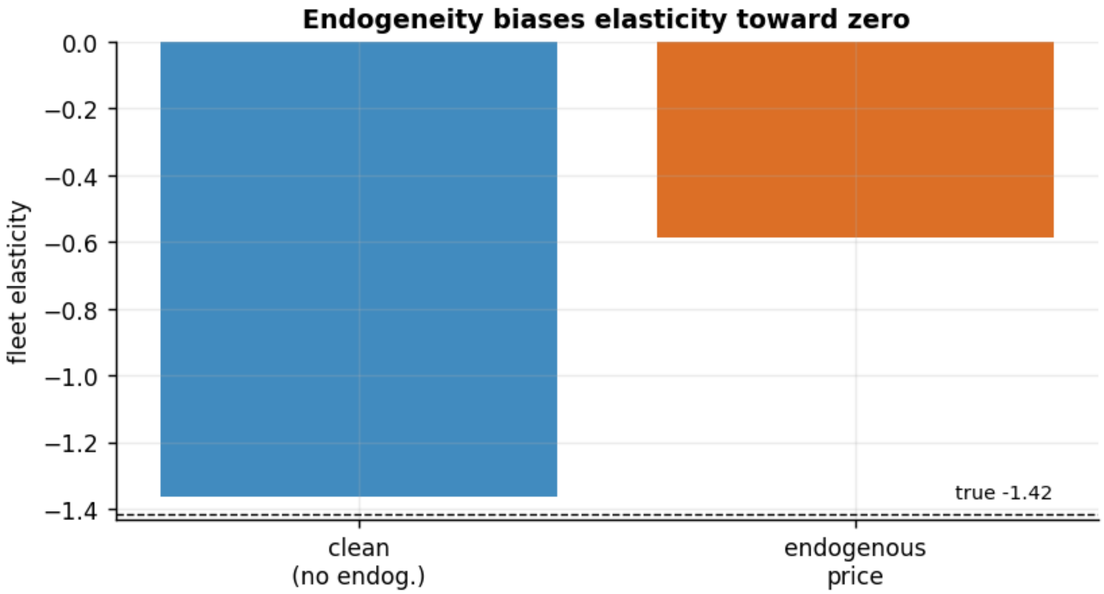
region	elasticity	P(elastic)	opt_price	action
region_00	-1.07	0.63	€43.4	controlled test
region_01	-1.72	1.00	€19.6	set price
region_02	-1.17	0.75	€38.4	controlled test
region_03	-1.20	0.88	€41.7	controlled test
region_04	-1.18	0.85	€45.5	controlled test
region_05	-1.29	0.97	€34.4	set price
region_06	-1.57	1.00	€22.2	set price
region_07	-1.65	1.00	€20.3	set price
region_08	-1.29	0.98	€35.1	set price
region_09	-1.17	0.85	€47.0	controlled test
region_10	-1.54	1.00	€22.7	set price
region_11	-1.50	1.00	€24.0	set price



1.6 7 · Caveats — what partial pooling does *not* fix

Partial pooling handles *noise*, not *bias*. If price is endogenous to an **unobserved** demand shock, adjusting for observed Z isn't enough and elasticity is biased **toward zero**. We demonstrate by switching the hidden shock on — and preview the IV cure.

```
clean fleet elasticity -1.36 (true -1.42)
endogenous fleet elasticity -0.59 (true -1.42) -> biased toward zero by +0.83
Fix = an instrument (a cost shifter) or a natural price experiment - see
notebook 11 (IV).
```



Other caveats. Log-log assumes *constant* elasticity within region (a local approximation if elasticity varies with price level); `lag(price)` would capture stockpiling dynamics (omitted for clarity); and shrinkage is a prior choice — too few regions or a mis-specified τ_β prior can over-pool, so always inspect the fleet-vs-region spread.